SCHOOL
SUPERIOR
AVERAGE
ARTS
AND DESIGN
POLYTECHNIC
FROM PORTO

ALP

WORKSHEET

TEACHING ACTIVITY

Algorithmics and Data Structures

COURSE UNIT

worksheet 07 – File handling

APPLICATION 1 – TEXT FILE

Create an application that allows you to manage a country file (***paises.txt***), with a structure like the one shown in the example: *country; continent*

```
paises - Bloco de notas
Ficheiro  Editar  Formatar  Ver  Ajuda
Portual;Europa
Espanha;Europa
França;Europa
Itália;Europa
Suécia;Europa
EUA;América
Brasil;América
Peru;América
Colombia;América
China;Asia
Japão;Asia
Tailandia;Asia
Malásia;Asia
```

Your app should include a menu with the following options:

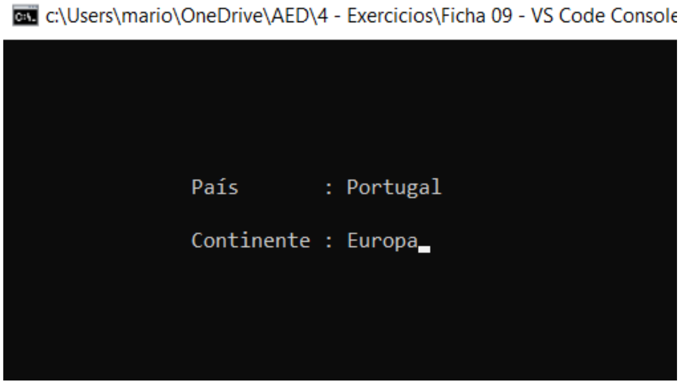
```
c:\Users\mario\OneDrive\AED\4 - Exercicios\Ficha

MENU
1 - Inserir Países
2 - Consulta Países
3 - Consulta por continente
4 - Consulta nº países
0 - Sair
        opção:
```

1- Inserir Países | Add Countries

Call a function names ***addCountry*()** that:

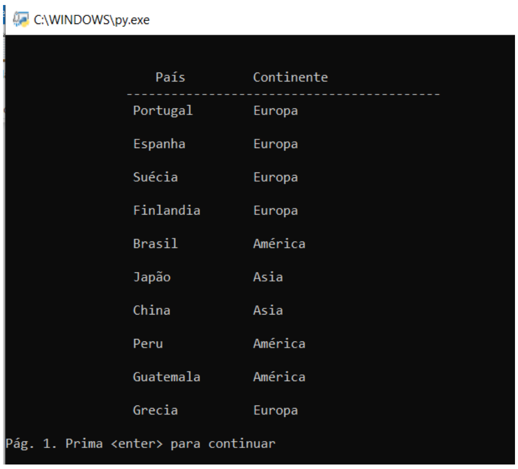
- It asks the user to enter a country and a continent;
- If the ***países.txt*** file does not exist, your application must create it, in a sub-folder ***files*** within your project folder.
- Check to see if the country already exists in the file. If there is, you must show a message to the user and <u>not duplicate</u> the country name in the file.

- If the country does not exist in the file, you must add it to the country file.



2- Consulta Países | View Countries

- You must call a function ***showCountries()*** that allows you to list all the countries in the países.txt file.
- If your file has more than 10 countries, please list 10 countries per page.



3- Consulta por continente | By Continent

- You must ask the user to indicate the continent.
- Next, you should call a function (which receives the continent indicated by the user) that allows the countries of that continent to be listed on file.

c:\Users\mario\OneDrive\AED\4 - Exercicios\Ficha 09 - VS Code

```
Continente: Europa_
```

c:\Users\mario\OneDrive\AED\4 - Exercicios\Ficha 09 - VS Code Console

```
   País          Continente
 ------------------------------------------
 Portual         Europa

 Espanha         Europa

 França          Europa

 Itália          Europa

 Suécia          Europa
```

4- Consulta nº Países por Continente | Number of Countries by Continent

- This option should call a function that prints the number of countries, on file, by *each continent

Application 2: Bynary File

1. Implement the **writeText(**text) function that receives a text (read, e.g. through an input) and saves it in a binary file, called *text.bin*. If the file does not exist, it must be created in subfolder *files*.

2. Implement the **readText**() function that reads the contents of the *./files/text.bin file* and returns the corresponding text. Print the result, returned by the function, on the console application.

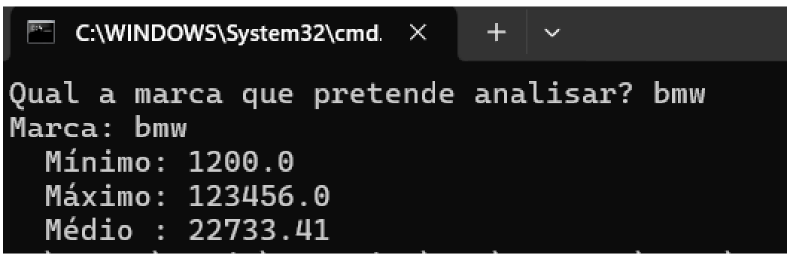
   Verify that the file exists and that no errors occur during its reading.

Note: the *path* and filename should be easily parameterized in your code (global variables), without the need to change the code of the functions requested above.

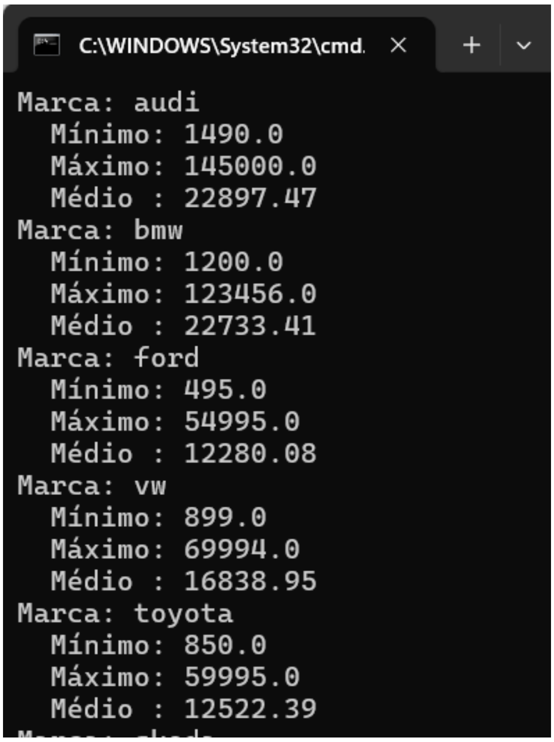
Application 3: CSV File

Open the ***carros_usados.csv*** file which is a *dataset* of data relating to the sale of cars from a German company (made available for free, like many other datasets, through the [https://www.kaggle.com/datasets platform](https://www.kaggle.com/datasets)).

3.1 It is intended that you implement a small application that can load the dataset data and create a "report" **with data from a specific car brand, indicated by the user**.



3.2 Implement now an application that can load the dataset data and create a "report" with **data from all brands in the dataset**, as illustrated below:

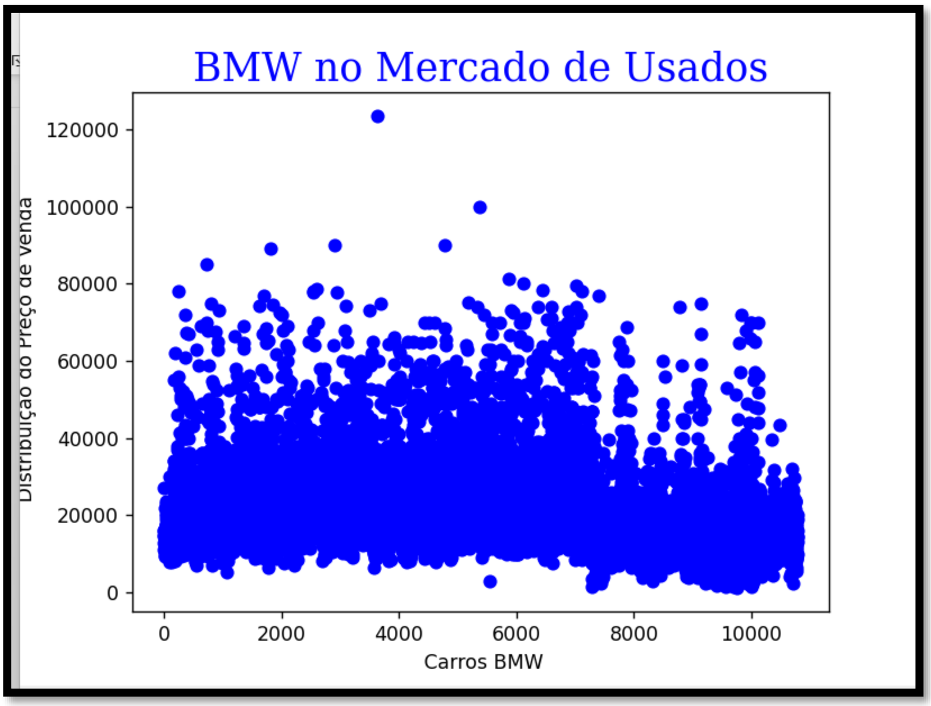


```
C:\WINDOWS\System32\cmd.
Marca: audi
  Mínimo: 1490.0
  Máximo: 145000.0
  Médio : 22897.47
Marca: bmw
  Mínimo: 1200.0
  Máximo: 123456.0
  Médio : 22733.41
Marca: ford
  Mínimo: 495.0
  Máximo: 54995.0
  Médio : 12280.08
Marca: vw
  Mínimo: 899.0
  Máximo: 69994.0
  Médio : 16838.95
Marca: toyota
  Mínimo: 850.0
  Máximo: 59995.0
  Médio : 12522.39
```

CHALLENGE:

At the end we cloud generate a graphic with the data collected in items 3.1 and 3.2.

NOTE: We will still discuss the Matplotlib library...



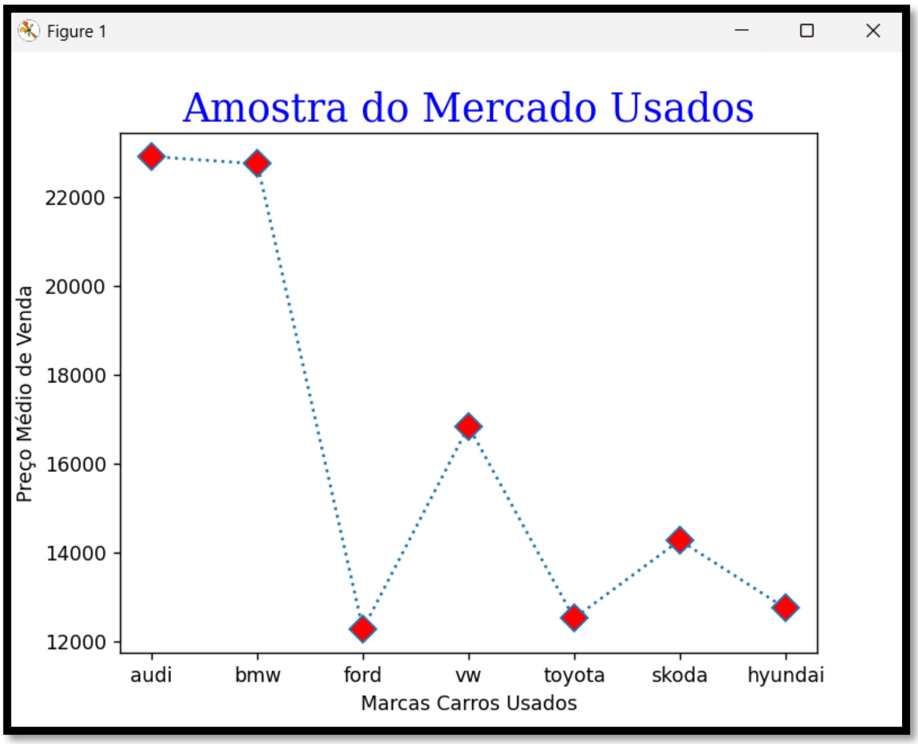


Figure 1
Amostra do Mercado Usados
22000
20000
18000
16000
14000
12000
Preço Médio de Venda
audi
bmw
ford
vw
toyota
skoda
hyundai
Marcas Carros Usados